

SOFTWARE REQUIREMENT SPECIFICATION

(SRS)

AI-Powered University Club Management System

Full-Scale Web Platform Edition — Version 1.0

Project Title	AI-Powered University Club Management System
Prepared By	Mehedi Ashik
Courses	Database Management Systems (DBMS) & Software Engineering (SWE)
Document Type	Software Requirement Specification — Week 2 Deliverable
Date	21 June 2026

Table of Contents

1. Introduction

1.1 Purpose

This Software Requirement Specification (SRS) document describes the functional and non-functional requirements for the AI-Powered University Club Management System. It is intended to serve as the foundation for design, development, and testing of the system, and as a reference for the project supervisor to evaluate the scope and completeness of the proposed solution.

1.2 Scope

The system is a full-scale, publicly accessible web platform that automates university club operations. It provides a public-facing website for guests to browse events and read club news, alongside a secure, role-based member portal for managing memberships, event registration, online payments, attendance, certificate generation, community discussion, and AI-assisted content creation.

The system supports four user roles — Super Admin, Club President, Member, and Guest — each with a clearly defined scope of access, as detailed in Section 3.

1.3 Intended Audience

- Project Supervisor — for evaluation of requirements coverage and design soundness
- Developer (Mehedi Ashik) — as a reference during implementation and testing
- Future maintainers — as documentation of original system intent

1.4 Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirement Specification
FR	Functional Requirement
NFR	Non-Functional Requirement
CRUD	Create, Read, Update, Delete
LLM	Large Language Model (used for AI content generation)
PG	Payment Gateway (SSLCommerz / bKash)
ERD	Entity-Relationship Diagram

2. Overall Description

2.1 Product Perspective

The system is a standalone, full-stack web application built on Django (Python) with a MySQL relational database. It is not a modification of an existing product, but a new system designed specifically for university club management, combining traditional administrative features with online payments and AI-assisted content generation.

2.2 Product Functions (Summary)

- Public website with landing page, event listings, blog, and contact form
- Role-based authentication and authorization (4 roles)
- Member management with application and approval workflow
- Event creation, public/member registration, and capacity management
- Online payment processing for event fees and membership dues
- QR-code based attendance tracking
- Automated PDF certificate generation with verification
- Blog and discussion forum for community engagement
- Role-specific analytics dashboards
- AI-assisted event description and blog draft generation
- In-app and email notifications

2.3 User Classes and Characteristics

Role	Characteristics
Super Admin	Full system access. Manages users, roles, site-wide settings, and payment oversight. Typically one or two trusted individuals (e.g. faculty advisor or lead developer).
Club President	Club-level management access. Creates and manages events, approves membership applications, publishes blog posts, and moderates the forum.
Member	Registered club member with a personal portal. Can register for events, pay fees, download certificates, and participate in the forum.
Guest	Unregistered public visitor. Can browse public pages, read the blog, and register/pay for public events without creating a full member account.

2.4 Operating Environment

- Server-side: Django 4.x (Python 3.11+), MySQL 8.0
- Client-side: Any modern web browser (Chrome, Firefox, Edge, Safari) on desktop or mobile
- Hosting: Render or Railway (cloud PaaS)

- Third-party services: SSLCommerz/bKash (payments), SMTP email provider, LLM API (Claude/OpenAI)

2.5 Design and Implementation Constraints

- Must use Django as the backend framework and MySQL as the database (course requirement)
- Payment processing limited to gateways supporting Bangladeshi transactions (SSLCommerz, bKash)
- AI features depend on third-party LLM API availability and usage quotas
- Development timeline constrained to an 11-week academic schedule

2.6 Assumptions and Dependencies

- Users have access to a stable internet connection
- The university provides no existing system this project must integrate with
- SSLCommerz/bKash sandbox credentials are available for development and testing

3. System Actors

The system defines four actors with a hierarchical permission structure: each role inherits the capabilities of the role below it, with additional privileges layered on top (Guest → Member → Club President → Super Admin).

Actor	Primary Goals
Guest	Explore club activities and optionally register/pay for public events without full membership.
Member	Participate fully in club life — register for events, track attendance history, collect certificates, engage in forum discussions.
Club President	Run day-to-day club operations — organize events, manage membership intake, and maintain community content.
Super Admin	Oversee the entire platform — user/role management, payment oversight, and system configuration.

4. Use Case Diagram

The diagram below illustrates all identified actors and their associated use cases, grouped by functional area. Generalization arrows indicate role inheritance — for example, a Club President can perform all Member-level actions in addition to their own.

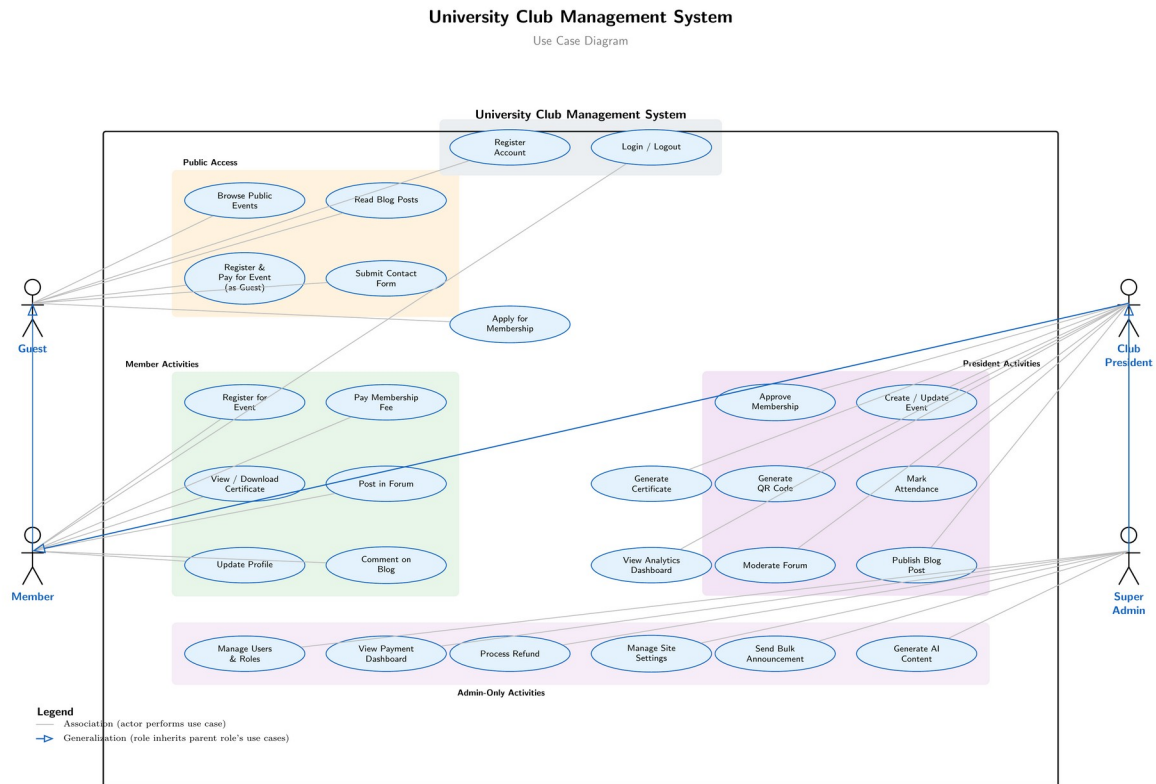


Figure 4.1 — Use Case Diagram for the University Club Management System

5. Functional Requirements

Each functional requirement below is mapped to its corresponding use case and responsible actor(s), ensuring full traceability to Section 4.

5.1 Authentication & User Management

ID	Requirement	Related Use Case / Actor
FR-01	The system shall allow a Guest to register a new account by providing a unique username, valid email address, and password.	Register Account — Guest
FR-02	The system shall verify the user's email address before activating the account.	Register Account — Guest
FR-03	The system shall allow a registered user to log in using their email/username and password, and to log out.	Login/Logout — Member, President, Admin
FR-04	The system shall allow login via Google OAuth as an alternative to email/password.	Login/Logout — Member, President, Admin
FR-05	The system shall allow a Super Admin to assign or change a user's role (Member, President, Admin).	Manage Users & Roles — Admin
FR-06	The system shall allow users to update their profile information and reset a forgotten password.	Update Profile — Member

5.2 Member Management

ID	Requirement	Related Use Case / Actor
FR-07	The system shall allow a Guest to submit a membership application with department, batch, and roll number.	Apply for Membership — Guest
FR-08	The system shall allow a Club President to review, approve, or reject pending membership applications.	Approve Membership — President
FR-09	The system shall provide a searchable, filterable member directory accessible to Members and above.	(supporting) — Member, President

5.3 Event Management

ID	Requirement	Related Use Case / Actor
FR-10	The system shall allow a Club President to create, update, and cancel events, specifying	Create/Update Event — President

ID	Requirement	Related Use Case / Actor
	title, description, category, date, venue, capacity, and fee.	
FR-11	The system shall allow events to be marked as public (visible to Guests) or member-only.	Create/Update Event — President
FR-12	The system shall allow a Member or Guest to register for an open event, subject to available capacity.	Register for Event — Member; Register & Pay (Guest) — Guest
FR-13	The system shall automatically close registration once an event reaches its capacity limit.	Register for Event — Member, Guest

5.4 Payment Processing

ID	Requirement	Related Use Case / Actor
FR-14	The system shall integrate with SSLCommerz/bKash to process event registration fees and membership dues online.	Pay Membership Fee; Register & Pay (Guest)
FR-15	The system shall record each transaction's amount, method, transaction ID, and status (Paid/Pending/Failed).	View Payment Dashboard — Admin
FR-16	The system shall generate a downloadable payment receipt (PDF) after a successful transaction.	(supporting) — Member, Guest
FR-17	The system shall allow a Super Admin to process refund requests.	Process Refund — Admin

5.5 Attendance Management

ID	Requirement	Related Use Case / Actor
FR-18	The system shall generate a unique QR code for each confirmed event registration.	Generate QR Code — President
FR-19	The system shall allow a Club President to mark attendance by scanning a member's QR code or via manual entry.	Mark Attendance — President
FR-20	The system shall provide an exportable attendance report per event and per member.	(supporting) — President, Admin

5.6 Certificate Generation

ID	Requirement	Related Use Case / Actor
FR-21	The system shall automatically generate a PDF	Generate Certificate — President

ID	Requirement	Related Use Case / Actor
	certificate for members marked present at a completed event.	
FR-22	Each certificate shall include a unique, verifiable certificate ID.	Generate Certificate — President
FR-23	The system shall allow a Member to view and download their certificates from their personal portal.	View/Download Certificate — Member

5.7 Blog & Forum

ID	Requirement	Related Use Case / Actor
FR-24	The system shall allow a Club President to create, publish, and archive blog posts using a rich text editor.	Publish Blog Post — President
FR-25	The system shall allow any registered user to comment on a published blog post.	Comment on Blog — Member
FR-26	The system shall allow a Member to create discussion threads and post replies in the forum.	Post in Forum — Member
FR-27	The system shall allow a Club President to pin, lock, or delete forum threads as a moderation action.	Moderate Forum — President

5.8 Dashboard & Analytics

ID	Requirement	Related Use Case / Actor
FR-28	The system shall display a role-specific analytics dashboard showing relevant statistics (e.g. member growth, event frequency, revenue) using charts.	View Analytics Dashboard — President, Admin

5.9 AI Features

ID	Requirement	Related Use Case / Actor
FR-29	The system shall generate a draft event description from a title, category, and date using an LLM API.	Generate AI Content — Admin/President
FR-30	The system shall generate a draft blog post from a given topic using an LLM API.	Generate AI Content — Admin/President
FR-31	All AI-generated content shall be editable by the user before being saved or published.	Generate AI Content — Admin/President

5.10 Notifications

ID	Requirement	Related Use Case / Actor
FR-32	The system shall send an in-app and/or email notification to a member upon successful event registration, payment confirmation, or certificate issuance.	(supporting) — Member
FR-33	The system shall allow a Super Admin to send a bulk announcement email to all members.	Send Bulk Announcement — Admin

5.11 Site Administration

ID	Requirement	Related Use Case / Actor
FR-34	The system shall allow a Guest to submit an inquiry via a public contact form.	Submit Contact Form — Guest
FR-35	The system shall allow a Super Admin to configure site-wide settings.	Manage Site Settings — Admin

6. Non-Functional Requirements

Non-functional requirements define the quality attributes of the system — how well it performs its functions, rather than what functions it performs.

Category	ID	Requirement
Performance	NFR-01	95% of page requests shall complete within 2 seconds under normal load (≤ 100 concurrent users).
Performance	NFR-02	The AI content generation feature shall return a result within 10 seconds, with a loading indicator shown to the user.
Security	NFR-03	All user passwords shall be stored as salted hashes (never in plain text), using Django's built-in password hashing.
Security	NFR-04	All data transmission shall be encrypted via HTTPS (TLS).
Security	NFR-05	The system shall implement CSRF protection on all state-changing forms (Django default).
Security	NFR-06	Role-based access control shall prevent users from accessing pages or actions outside their permission level.
Usability	NFR-07	The user interface shall be responsive and usable on screen widths from 360px (mobile) to 1920px (desktop).
Usability	NFR-08	Critical user actions (payment, registration) shall provide clear success/failure feedback.
Reliability	NFR-09	The system shall maintain 99% uptime during the academic semester, excluding scheduled maintenance.
Reliability	NFR-10	Payment transaction records shall never be lost, even if the payment gateway callback fails (reconciliation job required).
Scalability	NFR-11	The database schema shall be normalized to 3NF to minimize redundancy and support future growth in record volume.
Maintainability	NFR-12	The codebase shall follow Django's app-based modular structure, with each major feature isolated into its own app.
Compatibility	NFR-13	The system shall function correctly on the latest two major versions of Chrome, Firefox, Edge, and Safari.
Compliance	NFR-14	Payment processing shall comply with SSLCommerz/bKash's integration and data-handling guidelines.

7. Data Requirements

The system's data is organized into twelve core entities, normalized to Third Normal Form (3NF) to eliminate redundancy. The full Entity-Relationship Diagram is provided below; the complete schema with data types and constraints is available in docs/ER_Diagram.pdf and database/schema.sql.

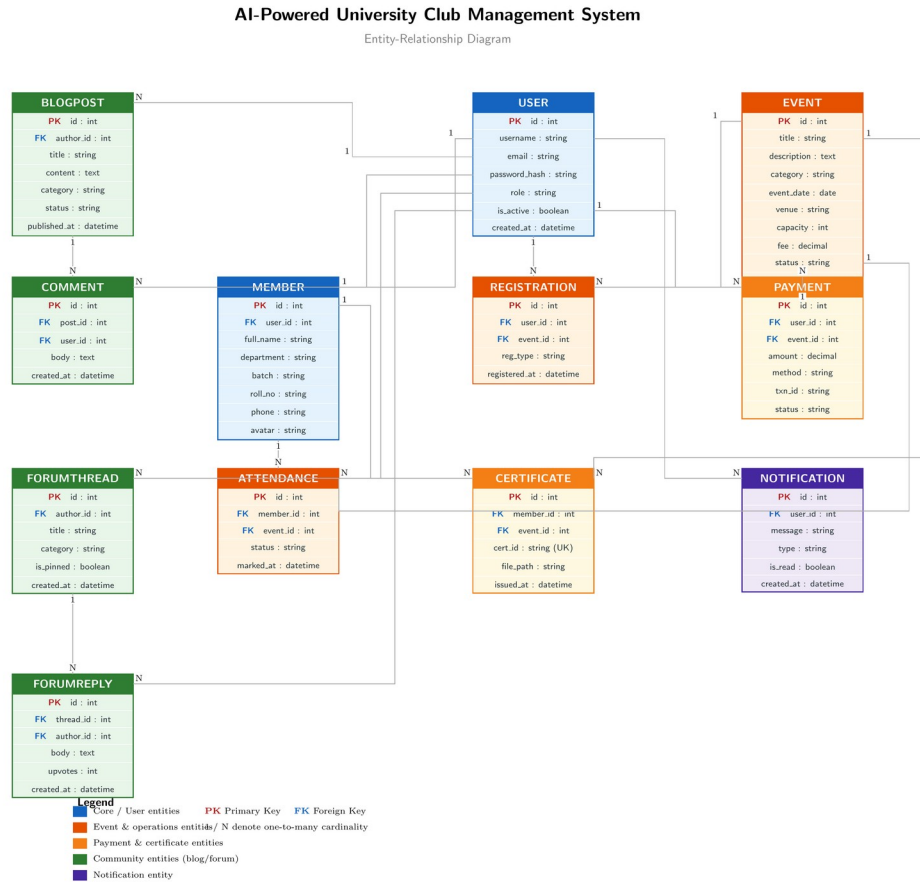


Figure 7.1 — Entity-Relationship Diagram

8. Appendix

8.1 Related Documents

- Project Roadmap — docs/roadmap.pdf
- Entity-Relationship Diagram — docs/ER_Diagram.pdf
- UML Class Diagram — docs/UML_Class_Diagram.pdf
- Use Case Diagram — docs/Use_Case_Diagram.pdf
- Database Schema (DDL) — database/schema.sql

8.2 Revision History

Version	Date	Description
1.0	21 Jun 2026	Initial SRS — Week 2 deliverable